TATA TECHNOLOGIES

TATA

TechVarsity
Center for lifelong learning

TechPULSE
Industry Academia Partnership

Tata Technologies Ltd.

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Plot No 25, Rajiv Gandhi Infotech Park, Hinjawadi, Pune, India – 411 057

TechPulse FY- 26

Course Name: Applied AI ML

Track: AI & ML

Course Level: Intermediate

Learning Objectives:

The students will be able to,

1. Understand the fundamentals of Artificial Intelligence and Machine Learning in an industrial and automotive context.
2. Differentiate between supervised, unsupervised, and reinforcement learning approaches.
3. Preprocess and analyze real-world automotive datasets using Python.
4. Build machine learning and deep learning models for predictive maintenance and vision-based applications.
5. Apply computer vision techniques for traffic sign recognition and object detection.
6. Understand the working principles of neural networks, CNNs, RNNs, and transfer learning.
7. Explore Generative AI and apply prompt engineering techniques for automotive diagnostics.

Course Outline:

Sl.No	Topics recommended	Theory Hours	LAB hours
UNIT 1 **Introduction to AI & ML**	1. Basics of AI and ML 2. Difference between AI, ML, and DL 3. Types of ML (Supervised, Unsupervised, Reinforcement) 4. Automotive use cases (predictive maintenance, driver monitoring) 5. Components of an AI system 6. Data collection and preprocessing 7. ML lifecycle overview 8. Model evaluation basics 9. Introduction to Python libraries (NumPy, Pandas, Scikit-	4	

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	learn) 10. Ethical AI in automotive systems		
UNIT 2 – Machine Learning & Applications	1. Understanding datasets and preprocessing 2. Handling missing values and outliers 3. Feature scaling and encoding 4. EDA using automotive datasets 5. Linear Regression for mileage prediction 6. Decision Trees and Random Forests 7. K-Means clustering for vehicle grouping 8. Principal Component Analysis (PCA) 9. Evaluating model performance (MAE, RMSE, R^2) 10. Case study: Car price prediction	8	4
UNIT 3 – Deep Learning and Computer Vision	1. Fundamentals of Neural Networks 2. Activation functions 3. Forward and backward propagation 4. CNN architecture 5. Image preprocessing for automotive datasets 6. Traffic sign recognition 7. Lane detection basics 8. Object detection overview (YOLO, SSD) 9. Overfitting and dropout 10. Confusion matrix and model validation	8	4
UNIT 4 AI fundamentals & application development	1. Math for AI (Linear Algebra, Calculus, Probability) 2. Feature selection and importance 3. SVM and Ensemble methods 4. Gradient Boosting and XGBoost 5. RNN and LSTM overview 6. Text data preprocessing 7. Transfer learning concepts 8. Model optimization 9. Deployment basics 10. Evaluation metrics (precision, recall, F1-score)	8	4
UNIT 5 - Generative AI, Prompt Engineering & MLOps Overview	**1**. Introduction to Generative AI 2. GPT models and their applications 3. Prompt Engineering basics 4. Structured prompts for automotive diagnostics 5. Overview of MLOps 6. CI/CD in ML pipelines 7. Model versioning and tracking 8. MLflow and Kubeflow introduction 9. Containerization using Docker 10. Responsible AI and deployment ethics	4	2
	TOTAL		

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Lab Statements :

Sr. No.	Lab Title	Description
1	ML Model for Car Mileage Estimation	Predict car mileage using regression and Python libraries.
2	Simulated Driving Agent Behavior	Build a self-driving agent using Python and genetic algorithms.
3	Data Cleaning & Preprocessing Techniques	Handle missing values, outliers, and scale features using Pandas and Scikit-learn.
4	Vehicle Price Prediction	Use regression models to estimate vehicle prices from structured data.
5	Predictive Maintenance from Sensor Logs	Classify component failures using sensor data and Python ML models.
6	Traffic Sign Classification using CNN	Train a CNN to recognize traffic signs using the GTSRB dataset.
7	Pedestrian Detection using OpenCV	Detect pedestrians using OpenCV's HOG + SVM method.
8	Sentiment Analysis using LSTM	Analyze vehicle feedback using LSTM-based sentiment classification.
9	Feature Importance Visualization	Visualize and interpret feature importance in automotive datasets.
10	MLOps Workflow Simulation	Build a CI/CD pipeline using MLflow and Docker for model deployment.

Prerequisite:

- **Basic programming skills in Python**
- **Familiarity with data structures and algorithms**
- **Introductory understanding of machine learning concepts**
- **Exposure to linear algebra, statistics, and probability**
- **Comfort with using Jupyter Notebooks or Google Colab**
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Books recommended:

Introduction to AI & Machine Learning — Munesh Chandra Trivedi & Ankit Srivastava

Python Machine Learning — Sebastian Raschka & Vahid Mirjalili

Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow — Aurélien Géron

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Software Requirements:

- Python 3.8+
- Jupyter Notebook / Google Colab
- NumPy
- Pandas
- Scikit-learn
- Matplotlib
- Seaborn
- TensorFlow / Keras
- OpenCV
- MLflow
- Docker
- GitHub

Evaluation:

1. In class assessment by college,
2. Final Assessment by Tata Technologies

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